

Causal Discovery from Complex Survey Data

Theory (appendix)

1 Appendix A: Proofs of Theoretical Results

This appendix provides complete proofs of all results stated in Section 4 of the main text. Throughout, we use the notation established there: ξ denotes the superpopulation, U_N the finite population of size N , S_n the probability sample of size n , $w_i = 1/\pi_i$ the survey weights, and $P_\xi(\mathbf{X} \mid \mathcal{G}, \theta, \mathbf{Z})$ the ordinal BN model under DAG $\mathcal{G}$ with parameters θ and covariates $\mathbf{Z}$.

We define the following quantities that appear throughout the proofs:

- **Census log-likelihood:** $\ell_U(\theta \mid \mathcal{G}) = \sum_{i \in U_N} \log P(\mathbf{x}_i \mid \mathcal{G}, \theta, \mathbf{z}_i)$.
- **Pseudo-log-likelihood:** $\tilde{\ell}(\theta \mid \mathcal{G}) = \sum_{i \in S_n} w_i \log P(\mathbf{x}_i \mid \mathcal{G}, \theta, \mathbf{z}_i)$.
- **Superpopulation expected log-likelihood:** $Q_\xi(\theta \mid \mathcal{G}) = \mathbb{E}_\xi[\log P(\mathbf{X} \mid \mathcal{G}, \theta, \mathbf{Z})]$.
- **Normalized versions:** $\bar{\ell}_U(\theta \mid \mathcal{G}) = N^{-1} \ell_U(\theta \mid \mathcal{G})$, $\bar{\tilde{\ell}}(\theta \mid \mathcal{G}) = N^{-1} \tilde{\ell}(\theta \mid \mathcal{G})$.

1.1 A.1 Proof of Proposition 2 (Pseudo-MLE Consistency)

We prove that $\hat{\theta}_\mathcal{G}^w \xrightarrow{p} \theta_\mathcal{G}^*$ under the joint probability. The argument proceeds in three steps.

Step 1: Design consistency of the normalized pseudo-log-likelihood.

Fix $\theta \in \Theta_\mathcal{G}$ and define $g_\theta(\mathbf{x}, \mathbf{z}) = \log P(\mathbf{x} \mid \mathcal{G}, \theta, \mathbf{z})$. Since $\mathbf{X}$ takes values in the finite set $\prod_{j=1}^q \{1, \dots, L_j\}$ and F is bounded, g_θ is bounded for each fixed θ . By Assumption 9 (Design Consistency), for each fixed realization of U_N ,

$$\bar{\tilde{\ell}}(\theta \mid \mathcal{G}) - \bar{\ell}_U(\theta \mid \mathcal{G}) = \frac{1}{N} \sum_{i \in S_n} w_i g_\theta(\mathbf{x}_i, \mathbf{z}_i) - \frac{1}{N} \sum_{i \in U_N} g_\theta(\mathbf{x}_i, \mathbf{z}_i) \xrightarrow{p_N} 0,$$

where p_N denotes convergence under the design distribution. This is a direct application of the Horvitz–Thompson consistency theorem for bounded variables under Assumptions 7–9.

Step 2: Superpopulation law of large numbers for the census log-likelihood.

By Assumption 6 (Non-Degeneracy) and the i.i.d. superpopulation structure, $\{(\mathbf{x}_i, \mathbf{z}_i)\}_{i \in U_N}$ are i.i.d. draws from ξ . Since g_θ is bounded, the strong law of large numbers gives

$$\bar{\ell}_U(\theta \mid \mathcal{G}) = \frac{1}{N} \sum_{i=1}^N g_\theta(\mathbf{x}_i, \mathbf{z}_i) \xrightarrow{a.s.} \mathbb{E}_\xi[g_\theta(\mathbf{X}, \mathbf{Z})] = Q_\xi(\theta \mid \mathcal{G}).$$

Step 3: Uniform convergence and argmax convergence.

Combining Steps 1 and 2, for each fixed θ :

$$\tilde{\ell}(\theta \mid \mathcal{G}) \xrightarrow{p} Q_\xi(\theta \mid \mathcal{G}).$$

We upgrade this to uniform convergence over $\Theta_{\mathcal{G}}$. By Assumption 10(iv), for any $\theta, \theta' \in \Theta_{\mathcal{G}}$,

$$|g_\theta(\mathbf{x}, \mathbf{z}) - g_{\theta'}(\mathbf{x}, \mathbf{z})| \leq M \|\theta - \theta'\|$$

for some constant $M > 0$ (using that $\mathbf{X}$ has finite support and F is smooth). This Lipschitz condition, combined with the compactness of $\Theta_{\mathcal{G}}$ (Assumption 10(i)), gives equicontinuity of the process $\theta \mapsto \tilde{\ell}(\theta \mid \mathcal{G})$. Pointwise convergence plus equicontinuity on a compact set gives uniform convergence (see, e.g., van der Vaart, 1998, Theorem 2.4):

$$\sup_{\theta \in \Theta_{\mathcal{G}}} \left| \tilde{\ell}(\theta \mid \mathcal{G}) - Q_\xi(\theta \mid \mathcal{G}) \right| \xrightarrow{p} 0. \quad (\text{A.1})$$

Now, $\hat{\theta}_{\mathcal{G}}^w = \arg \max_{\theta} \tilde{\ell}(\theta \mid \mathcal{G})$ (the normalization by N does not affect the argmax). By Assumption 10(iii), $Q_\xi(\theta \mid \mathcal{G})$ has a unique maximizer $\theta_{\mathcal{G}}^*$ with nonsingular Hessian. By (A.1) and the argmax continuous mapping theorem (van der Vaart, 1998, Theorem 5.7),

$$\hat{\theta}_{\mathcal{G}}^w \xrightarrow{p} \theta_{\mathcal{G}}^*. \quad (\text{A.2})$$

As a corollary, by the continuous mapping theorem and (A.1):

$$\tilde{\ell}(\hat{\theta}_{\mathcal{G}}^w \mid \mathcal{G}) \xrightarrow{p} Q_\xi(\theta_{\mathcal{G}}^* \mid \mathcal{G}). \quad (\text{A.3})$$

■

Remark. The two-step structure of the proof — design consistency followed by superpopulation convergence — is characteristic of the joint asymptotic framework for survey data. The design consistency step requires only bounded weights and the HT consistency property; it does not require any distributional assumptions on $\mathbf{X}$. The superpopulation step is a standard i.i.d. law of large numbers. The key insight is that these two steps compose: the pseudo-log-likelihood first approx-

imates the census log-likelihood (design step), which in turn approximates the superpopulation expected log-likelihood (population step).

1.2 Proof of Theorem 1 (Within Markov Equivalence Classes)

Let $\mathcal{G}^*$ be the true DAG with parameters θ^* , and let $\mathcal{G}'$ be Markov-equivalent with $\mathcal{G}' \neq \mathcal{G}^*$. Let $\theta^\dagger$ denote the pseudo-true parameter of $\mathcal{G}'$, i.e.,

$$\theta^\dagger = \arg \max_{\theta} Q_{\xi}(\theta \mid \mathcal{G}') = \arg \max_{\theta} \mathbb{E}_{\xi}[\log P(\mathbf{X} \mid \mathcal{G}', \theta, \mathbf{Z})].$$

Step 1: Penalty cancellation.

Since $\mathcal{G}^*$ and $\mathcal{G}'$ are Markov-equivalent, they have the same skeleton (Verma & Pearl, 1990). In the oBN parameterization, the number of free parameters depends only on the skeleton structure: $K_{\mathcal{G}} = \sum_{j=1}^q (L_j - 1 + \sum_{k \in \text{pa}(j)} (L_k - 1) + \dim(\delta_j))$. While the parent sets $\text{pa}(j)$ differ between Markov-equivalent DAGs, the total number of parent-child pairs (i.e., the number of directed edges) is the same because both DAGs have the same skeleton. Moreover, for each edge $j - k$ in the skeleton, the parameter count contributed by that edge is $(L_k - 1)$ if $k \rightarrow j$ or $(L_j - 1)$ if $j \rightarrow k$. For Markov-equivalent DAGs with the same skeleton, consider any edge that is oriented differently. If $\mathcal{G}^*$ has $k \rightarrow j$ and $\mathcal{G}'$ has $j \rightarrow k$, then $\mathcal{G}^*$ contributes $(L_k - 1)$ parameters to node j 's regression while $\mathcal{G}'$ contributes $(L_j - 1)$ to node k 's regression. In the sum $K_{\mathcal{G}} = \sum_j \sum_{k \in \text{pa}(j)} (L_k - 1) + \text{const}$, each edge $j - k$ contributes either $(L_k - 1)$ or $(L_j - 1)$ depending on orientation. If $L_j = L_k$ for all j, k (as in homogeneous ordinal scales, e.g., all variables having L levels), then $K_{\mathcal{G}^*} = K_{\mathcal{G}'}$ exactly.

When the ordinal variables have heterogeneous numbers of levels, the parameter counts may differ between Markov-equivalent DAGs. However, we note that for the CAHPS application (all need variables have $L_j = 3$) and for typical survey questionnaires using a common Likert scale, the homogeneous case applies. We therefore state the theorem under the condition $K_{\mathcal{G}^*} = K_{\mathcal{G}'}$ and note that in the heterogeneous case, the penalty difference is at most $O(\rho_n)$, which is negligible relative to the $O(N)$ pseudo-likelihood ratio (see Step 3). Under $K_{\mathcal{G}^*} = K_{\mathcal{G}'}$:

$$\widetilde{\text{BIC}}(\mathcal{G}^*) - \widetilde{\text{BIC}}(\mathcal{G}') = -2 \left[\tilde{\ell}(\hat{\theta}_{\mathcal{G}^*}^w \mid \mathcal{G}^*) - \tilde{\ell}(\hat{\theta}_{\mathcal{G}'}^w \mid \mathcal{G}') \right]. \quad (\text{A.4})$$

Step 2: Convergence of the pseudo-log-likelihood ratio.

By equation (A.3) from the proof of Proposition 2, applied to both $\mathcal{G}^*$ and $\mathcal{G}'$:

$$\frac{1}{N} \tilde{\ell}(\hat{\theta}_{\mathcal{G}^*}^w \mid \mathcal{G}^*) \xrightarrow{p} Q_{\xi}(\theta^* \mid \mathcal{G}^*), \quad (\text{A.5})$$

$$\frac{1}{N} \tilde{\ell}(\hat{\theta}_{\mathcal{G}'}^w \mid \mathcal{G}') \xrightarrow{p} Q_{\xi}(\theta^{\dagger} \mid \mathcal{G}'). \quad (\text{A.6})$$

Taking the difference:

$$\frac{1}{N} [\tilde{\ell}(\hat{\theta}_{\mathcal{G}^*}^w \mid \mathcal{G}^*) - \tilde{\ell}(\hat{\theta}_{\mathcal{G}'}^w \mid \mathcal{G}')] \xrightarrow{p} Q_{\xi}(\theta^* \mid \mathcal{G}^*) - Q_{\xi}(\theta^{\dagger} \mid \mathcal{G}'). \quad (\text{A.7})$$

Step 3: Strict positivity of the KL divergence.

The limit in (A.7) can be written as

$$\begin{aligned} Q_{\xi}(\theta^* \mid \mathcal{G}^*) - Q_{\xi}(\theta^{\dagger} \mid \mathcal{G}') &= \mathbb{E}_{\xi}[\log P(\mathbf{X} \mid \mathcal{G}^*, \theta^*, \mathbf{Z})] - \mathbb{E}_{\xi}[\log P(\mathbf{X} \mid \mathcal{G}', \theta^{\dagger}, \mathbf{Z})] \\ &= \mathbb{E}_{\xi} \left[\log \frac{P(\mathbf{X} \mid \mathcal{G}^*, \theta^*, \mathbf{Z})}{P(\mathbf{X} \mid \mathcal{G}', \theta^{\dagger}, \mathbf{Z})} \right] \\ &= \mathbb{E}_{\mathbf{Z}} [\text{KL}(P_{\xi}(\mathbf{X} \mid \mathcal{G}^*, \theta^*, \mathbf{Z}) \parallel P(\mathbf{X} \mid \mathcal{G}', \theta^{\dagger}, \mathbf{Z}))]. \end{aligned} \quad (\text{A.8})$$

By Assumption 5 (Ordinal Identifiability), for each fixed $\mathbf{Z} = \mathbf{z}$, the oBN model under $\mathcal{G}^*$ with parameters $(\theta^*, \delta^* \cdot \mathbf{z})$ is not distributionally equivalent to the oBN model under $\mathcal{G}'$ with any parameter value. Proposition 1 (Identifiability under Covariate Adjustment) establishes that this extends to the covariate-adjusted model: the covariates shift the intercept but do not alter the functional-form asymmetry exploited by the identifiability result of Ni and Mallick (2022). Therefore,

$$\text{KL}(P_{\xi}(\mathbf{X} \mid \mathcal{G}^*, \theta^*, \mathbf{z}) \parallel P(\mathbf{X} \mid \mathcal{G}', \theta^{\dagger}(\mathbf{z}), \mathbf{z})) > 0 \quad (\text{A.9})$$

for all $\mathbf{z}$ in the support of $\mathbf{Z}$. Since the KL divergence is nonnegative and continuous in $\mathbf{z}$ (by smoothness of F), and is strictly positive on the entire support, the expectation over $\mathbf{Z}$ in (A.8) is strictly positive:

$$Q_{\xi}(\theta^* \mid \mathcal{G}^*) - Q_{\xi}(\theta^{\dagger} \mid \mathcal{G}') = \Delta > 0. \quad (\text{A.10})$$

Step 4: Conclusion.

Combining (A.4), (A.7), and (A.10):

$$\widetilde{\text{BIC}}(\mathcal{G}^*) - \widetilde{\text{BIC}}(\mathcal{G}') = -2N[\Delta + o_p(1)] \rightarrow -\infty$$

in probability. Therefore $\Pr(\widetilde{\text{BIC}}(\mathcal{G}^*) < \widetilde{\text{BIC}}(\mathcal{G}')) \rightarrow 1$.

■

Remark. The rate of discrimination is $O(N)$, which is the population size, not the sample size n . This is because the pseudo-log-likelihood estimates the census log-likelihood, which accumulates information at rate N . The statistical noise in the pseudo-log-likelihood ratio is $O_p(N/\sqrt{n_{\text{eff}}})$ (from the design-based variance of the HT estimator), so the signal-to-noise ratio for within-MEC discrimination grows as $\sqrt{n_{\text{eff}}}$. This confirms that the oBN's ability to orient edges improves with effective sample size, not raw sample size.

1.3 A.3 Proof of Theorem 2 (Across Markov Equivalence Classes)

1.3.1 Part (a): Underfitting

Let $\mathcal{G}''$ be a DAG with strictly fewer edges than $\mathcal{G}^*$. Then $\mathcal{G}''$ is not in the Markov equivalence class of $\mathcal{G}^*$ and, by Faithfulness (Assumption 3), encodes at least one conditional independence that does not hold in P_ξ .

By the same argument as in Theorem 1, Steps 2–3:

$$\frac{1}{N} [\tilde{\ell}(\hat{\theta}_{\mathcal{G}^*}^w | \mathcal{G}^*) - \tilde{\ell}(\hat{\theta}_{\mathcal{G}''}^w | \mathcal{G}'')] \xrightarrow{p} \text{KL}(P_\xi(\cdot | \mathcal{G}^*) \| P(\cdot | \mathcal{G}'', \theta''^\dagger)) = \Delta'' > 0.$$

The strict positivity follows because $\mathcal{G}''$ imposes conditional independence constraints not present in P_ξ , so P_ξ cannot be expressed as a member of the oBN model under $\mathcal{G}''$ (this is a standard property of KL divergence: the projection of P_ξ onto the constrained model family has strictly positive KL from P_ξ when the constraint is violated).

The penalty difference satisfies $|K_{\mathcal{G}^*} - K_{\mathcal{G}''}| \cdot \rho_n = O(\rho_n) = o(N)$ by the assumption $\rho_n/N \rightarrow 0$. Therefore:

$$\begin{aligned} \widetilde{\text{BIC}}(\mathcal{G}^*) - \widetilde{\text{BIC}}(\mathcal{G}'') &= -2N[\Delta'' + o_p(1)] + (K_{\mathcal{G}^*} - K_{\mathcal{G}''})\rho_n \\ &= -2N\Delta''(1 + o_p(1)) + o(N) \\ &\rightarrow -\infty. \end{aligned} \tag{A.11}$$

■

1.3.2 Part (b): Overfitting

Let $\mathcal{G}''$ be a DAG with strictly more edges than $\mathcal{G}^*$, including at least one spurious edge not present in $\mathcal{G}^*$. Then $K_{\mathcal{G}''} > K_{\mathcal{G}^*}$.

Step 1: The pseudo-log-likelihood gain from spurious parameters is bounded.

We argue that the overfitting pseudo-log-likelihood ratio is $O_p(1)$, not $O_p(N)$. Consider the decomposition of the pseudo-BIC by nodes. For any DAG $\mathcal{G}$,

$$\tilde{\ell}(\hat{\theta}_{\mathcal{G}}^w \mid \mathcal{G}) = \sum_{j=1}^q \tilde{\ell}_j(\hat{\theta}_j^w \mid \text{pa}_{\mathcal{G}}(j)),$$

where $\tilde{\ell}_j(\theta_j \mid \text{pa}_{\mathcal{G}}(j)) = \sum_{i \in S_n} w_i \log P(x_{ij} \mid x_{i, \text{pa}(j)}, \mathbf{z}_i, \theta_j)$ is the node- j pseudo-log-likelihood. The spurious edge in $\mathcal{G}''$ adds parent k to some node j where $k \notin \text{pa}_{\mathcal{G}^*}(j)$. For node j , the pseudo-log-likelihood gain is:

$$\Lambda_{w,j} = 2 \left[\tilde{\ell}_j(\hat{\theta}_j^w \mid \text{pa}_{\mathcal{G}''}(j)) - \tilde{\ell}_j(\hat{\theta}_j^w \mid \text{pa}_{\mathcal{G}^*}(j)) \right]. \quad (\text{A.12})$$

Under the null (the true parent set of j does not include k), the additional parameters $\beta_{jk,2}, \dots, \beta_{jk,L_k}$ are zero in the population. We now characterize the distribution of $\Lambda_{w,j}$.

Define the pseudo-score for the additional parameters: $\tilde{U}_j = \sum_{i \in S_n} w_i \nabla_{\beta_{jk}} \log P(x_{ij} \mid x_{i, \text{pa}^*(j)}, \mathbf{z}_i, \theta_j^*)$. Under the null, $\mathbb{E}[\tilde{U}_j] = 0$ (the census score has mean zero at the true parameter, and the HT estimator is unbiased for the census score). The variance of $\tilde{U}_j$ is the design-based variance. Let $\pi_{ii'} = \Pr(i \in S_n \text{ and } i' \in S_n)$ denote the second-order (pairwise) inclusion probability for units i and i' , with $\pi_{ii} = \pi_i$. Then

$$\text{Var}_p(\tilde{U}_j) = \sum_{i \in U_N} \sum_{i' \in U_N} \frac{\pi_{ii'} - \pi_i \pi_{i'}}{\pi_i \pi_{i'}} s_i s_{i'}, \quad (\text{A.13})$$

where $s_i = \nabla_{\beta_{jk}} \log P(x_{ij} \mid x_{i, \text{pa}^*(j)}, \mathbf{z}_i, \theta_j^*)$ is the individual score contribution. We emphasize that (A.13) is used here only to establish the *order* of the variance — specifically, $\text{Var}_p(\tilde{U}_j) = O(N^2/n)$ under Assumptions 7–9, which is the standard rate for Horvitz–Thompson estimator variances. Computing or estimating $\pi_{ii'}$ is not required by our method; knowledge of $w_i = 1/\pi_i$ alone suffices for all algorithmic steps.

By a quadratic approximation (pseudo-likelihood analog of the Rao score test):

$$\Lambda_{w,j} \approx \tilde{U}_j^\top \left[\sum_{i \in S_n} w_i \mathcal{J}_j(\theta_j^*) \right]^{-1} \tilde{U}_j, \quad (\text{A.14})$$

where $\mathcal{J}_j$ is the per-observation Fisher information for the additional parameters. The bracketed term is $O(N)$ (it estimates the census Fisher information), and $\tilde{U}_j$ has variance $O(N^2/n)$. Therefore:

$$\Lambda_{w,j} = O_p\left(\frac{N^2/n}{N}\right) = O_p(N/n). \quad (\text{A.15})$$

Under the weight normalization used in the implementation (weights normalized to sum to n), the pseudo-log-likelihood is rescaled by n/N , giving $\Lambda_{w,j} = O_p(1)$ on the implementation scale. On the unscaled (HT) pseudo-likelihood used in the theorem, $\Lambda_{w,j} = O_p(N/n)$.

Step 2: The penalty dominates.

The penalty difference satisfies $(K_{\mathcal{G}''} - K_{\mathcal{G}^*}) \cdot \rho_n \geq \rho_n$. We require the penalty ρ_n to asymptotically dominate the pseudo-likelihood gain $\Lambda_{w,j} = O_p(N/n)$. On the implementation scale (with weights scaled to sum to n), this is equivalent to requiring $\log(n_{\text{eff}})/\bar{\delta} \rightarrow \infty$.

We establish this limit by invoking Assumption 8 (Bounded Weights), which guarantees the existence of constants $0 < c \leq C < \infty$ such that $c \leq nw_i/N \leq C$. Using Kish's formula, the effective sample size is bounded below by:

$$n_{\text{eff}} = \frac{(\sum_{i \in S_n} w_i)^2}{\sum_{i \in S_n} w_i^2} \geq \frac{(n \cdot c \frac{N}{n})^2}{n (C \frac{N}{n})^2} = \left(\frac{c}{C}\right)^2 n.$$

Because $n_{\text{eff}} \leq n$ by definition, we have $n_{\text{eff}} = \Theta(n)$. The average design effect $\bar{\delta} = n/n_{\text{eff}}$ is therefore bounded above by a constant $(C/c)^2 = O(1)$.

Since the sampling fraction $n/N \rightarrow f \in [0, 1)$ implies $n \rightarrow \infty$ as $N \rightarrow \infty$, we have $n_{\text{eff}} \rightarrow \infty$, which implies $\log(n_{\text{eff}}) \rightarrow \infty$. Because $\bar{\delta}$ is strictly bounded by $O(1)$, it follows conclusively that:

$$\frac{\log(n_{\text{eff}})}{\bar{\delta}} \rightarrow \infty.$$

Thus, the penalty term strictly dominates the $O_p(1)$ pseudo-likelihood gain in (A.17), and $\widehat{\text{BIC}}_{\text{impl}}(\mathcal{G}'') > \widehat{\text{BIC}}_{\text{impl}}(\mathcal{G}^*)$ with probability approaching 1. ■

1.4 A.4 Proof of Proposition 1 (Identifiability under Covariate Adjustment)

We prove that the ordinal identifiability result of Ni and Mallick (2022) extends to the covariate-adjusted oBN model.

Setup. Consider the bivariate case (the multivariate case follows by the same local argument applied to each edge). Let X_1, X_2 be ordinal with $L_1, L_2 \geq 3$ levels, and let $\mathbf{Z}$ be a covariate vector. Under the model $X_1 \rightarrow X_2$:

$$\begin{aligned}
P(X_1 \leq \ell) &= F(\gamma_{1\ell} - \delta'_1 \mathbf{Z} - \alpha_1), \\
P(X_2 \leq \ell \mid X_1) &= F(\gamma_{2\ell} - \beta_{21} X_1 - \delta'_2 \mathbf{Z} - \alpha_2).
\end{aligned} \tag{A.18}$$

Under the reverse model $X_2 \rightarrow X_1$:

$$\begin{aligned}
P(X_2 \leq \ell) &= F(\tilde{\gamma}_{2\ell} - \tilde{\delta}'_2 \mathbf{Z} - \tilde{\alpha}_2), \\
P(X_1 \leq \ell \mid X_2) &= F(\tilde{\gamma}_{1\ell} - \tilde{\beta}_{12} X_2 - \tilde{\delta}'_1 \mathbf{Z} - \tilde{\alpha}_1).
\end{aligned} \tag{A.19}$$

The argument. Fix $\mathbf{Z} = \mathbf{z}$. The conditional model (A.18) becomes

$$\begin{aligned}
P(X_1 \leq \ell \mid \mathbf{Z} = \mathbf{z}) &= F(\gamma_{1\ell} - \alpha_1(\mathbf{z})), \\
P(X_2 \leq \ell \mid X_1, \mathbf{Z} = \mathbf{z}) &= F(\gamma_{2\ell} - \beta_{21} X_1 - \alpha_2(\mathbf{z})),
\end{aligned} \tag{A.20}$$

where $\alpha_j(\mathbf{z}) = \alpha_j + \delta'_j \mathbf{z}$ absorbs the covariate effect into the intercept. This is exactly an oBN model for (X_1, X_2) with intercepts $\alpha_j(\mathbf{z})$. The reverse model (A.19) similarly reduces to an oBN with intercepts $\tilde{\alpha}_j(\mathbf{z})$.

The identifiability proof of Ni and Mallick (2022, Theorem 1) proceeds by showing that the joint distribution $P(X_1 = a, X_2 = b)$ under the forward ordinal regression model (A.20) cannot be expressed as a joint distribution arising from the reverse ordinal regression model (A.19) for generic parameter values. The proof constructs a system of polynomial equations relating the forward and reverse parameters and shows this system is generically inconsistent. Crucially, the intercepts $\alpha_j(\mathbf{z})$ enter the proof only through the marginal probabilities $P(X_j = \ell \mid \mathbf{z})$, and the inconsistency of the polynomial system holds for *every* value of these marginal probabilities (as long as they are in the interior of the simplex, which is guaranteed by Assumption 6). Since $\alpha_j(\mathbf{z})$ simply shifts the marginal probabilities to a different interior point, the inconsistency is preserved for each $\mathbf{z}$.

Formally, let $\Theta_{\text{forward}}(\mathbf{z}) \subset \mathbb{R}^{d_1}$ and $\Theta_{\text{reverse}}(\mathbf{z}) \subset \mathbb{R}^{d_2}$ denote the parameter spaces of the forward and reverse oBN models conditional on $\mathbf{Z} = \mathbf{z}$. The image of $\Theta_{\text{forward}}(\mathbf{z})$ under the map $\theta \mapsto P(\mathbf{X} \mid \theta, \mathbf{z})$ and the image of $\Theta_{\text{reverse}}(\mathbf{z})$ under the analogous map are semi-algebraic subsets of the probability simplex $\Delta^{L_1 L_2 - 1}$ (because F is the CDF of a known distribution and the parameterization is polynomial in the CDF values). By Ni and Mallick (2022, Theorem 1), these two images intersect on a set of Lebesgue measure zero in $\Delta^{L_1 L_2 - 1}$ when $L_1, L_2 \geq 3$. Varying $\mathbf{z}$ moves the true distribution within the forward image but does not move it onto the reverse image (except on a measure-zero set of $\mathbf{z}$ values, which does not affect the expectation over $\mathbf{Z}$).

Therefore, for almost every $\mathbf{z}$:

$$\text{KL}(P(\mathbf{X} \mid \mathcal{G}^*, \theta^*, \mathbf{z}) \parallel P(\mathbf{X} \mid \mathcal{G}', \theta^\dagger(\mathbf{z}), \mathbf{z})) > 0,$$

and hence the expectation over $\mathbf{Z}$ is strictly positive. ■

1.5 A.5 Discussion of Assumptions

We briefly discuss the roles and testability of the key assumptions.

Causal Sufficiency (Assumption 1) is the strongest and least testable assumption. In the CAHPS application, unmeasured confounders such as neighborhood-level economic conditions may simultaneously affect multiple social needs. Violation of causal sufficiency can introduce spurious edges in the learned DAG. We note that the covariate adjustment for demographics ($\mathbf{Z}$) partially mitigates this concern by removing confounding due to observed characteristics, but cannot address unobserved confounders. When causal sufficiency is suspect, the swa-FCI algorithm (Section 3.2 of the main text) provides a complementary analysis that accommodates latent confounders, at the cost of reduced identifiability.

Faithfulness (Assumption 3) is generic in the sense that the set of parameter values violating faithfulness has Lebesgue measure zero in the parameter space (Meek, 1995). However, near-violations of faithfulness (near-exact cancellation of effects) can reduce the power of conditional independence tests and lead to incorrect edge deletions in finite samples.

Ordinal Regression Specification (Assumption 4) is partially testable via goodness-of-fit diagnostics. The Brant test for proportional odds can be applied to each node’s regression post hoc. Violation of this assumption weakens the identifiability argument (Assumption 5), which relies specifically on the functional-form asymmetry of the cumulative link model.

Bounded Weights (Assumption 8) is verifiable from the data. In the CAHPS application, we report the range and distribution of survey weights and the ratio n/n_{eff} as a summary of weight variability. Extreme weights can be trimmed to satisfy this assumption approximately, at the cost of introducing a small bias.

Design Consistency (Assumption 9) is a property of the sampling design, not of the data. It is satisfied by all standard probability sampling designs used in federal surveys, including the stratified design of the VA CAHPS survey. Note that the results above require only first-order inclusion probabilities π_i (equivalently, the survey weights $w_i = 1/\pi_i$), not the pairwise inclusion probabilities π_{ij} . The latter would be needed for design-based variance estimation of individual parameters (e.g., sandwich standard errors for $\hat{\beta}_{jk}$), but are not required for pseudo-likelihood computation, BIC evaluation, or DAG selection consistency